

Separation methods
and instruments

* To make carbon dioxide: Place calcium carbonate in a flask and add dilute hydrochloric acid
 $\rightarrow \text{CaCO}_3(s) + 2\text{HCl}(aq) \rightarrow \text{CaCl}_2(aq) + \text{H}_2\text{O}(l) + \text{CO}_2(g)$

* To make hydrogen gas: Place pieces of zinc in a flask and add dilute hydrochloric acid.
 $\rightarrow \text{Zn}(s) + 2\text{HCl}(aq) \rightarrow \text{ZnCl}_2(aq) + \text{H}_2(g)$

* To make oxygen gas: Place manganese (IV) oxide as a catalyst in a flask, and add hydrogen peroxide
 $\rightarrow 2\text{H}_2\text{O}_2(aq) \rightarrow 2\text{H}_2\text{O}(l) + \text{O}_2(g)$

Collecting gases:

1- Upward displacement of air. Used when the gas is heavier than air.

Example: Carbon dioxide, sulfur dioxide, Hydrogen chloride.

2- Downward displacement of air. Used when the gas is lighter than air

Example: Ammonia, Hydrogen.

3- Over the water. Used when the gas is sparingly soluble in water.

Example: Carbon dioxide, Hydrogen, oxygen

4 Gas syringe. Used when you want to measure the volume of the gas accurately.

Separation methods:

Distillation: A way to separate substances from a liquid mixture, by boiling then condensing.

- Simple distillation: A way to obtain the solvent from a solution, leaving the solute behind.

1- Heat the solution in the flask. As it boils, water vapour rises into the condenser, leaving the solute behind.

2- The condenser is cold, so the vapour will condense back to water.

3- The water drips into a beaker. It is called distilled water; it is almost pure.

- Fractional distillation: Used to separate a mixture of liquids from each other by making use of their boiling points. It is like the apparatus used in simple distillation, but with a glass column filled with glass beads, and a thermometer.

Ex: Separate ethanol and water. ethanol boiling point 78°C , water: 100°C

1- Heat the mixture in the flask. At about 78°C , the ethanol begins to boil. Some of the water begins to evaporate too, so a mixture of water and ethanol vapour rises up the glass column.

2- The vapours condense on the flask beads in the column, making the beads heat up.

3- When the beads reach 78°C , ethanol vapour no longer condenses on them. Only water vapour does. So water drips back into the flask, while the ethanol vapour goes into the condenser.

4- It condenses there, pure liquid ethanol drips into the beaker.

5- Eventually, the thermometer reading rises above 78°C , a sign that all the ethanol has evaporated so you can stop heating.

Chromatography: Can be used to:

1- Separate substances from a mixture

2- Check the purity of a substance

3- Identify substances in a mixture.

1- Separating substances from a mixture:

* A thick line is drawn on a piece of chromatography paper using a black marker pen. The paper is stood in a beaker with little ethanol, and the line must be above the ethanol. The ethanol rises up the paper, dissolving the ink on the way. Bands of different colors appear.

$\rightarrow$ Ethanol rises up the paper by capillary action. When it reaches the ink, the ethanol dissolves the substances in the ink, and they travel up the paper with it. Even though they are all dissolved, each substance travels at a different speed, as it depends on their attraction to the ethanol and the paper. The substance that is most soluble in ethanol will travel fastest. The result is that all the substances separate. The paper showing a separated substance is called chromatogram

When you want to collect the separate substances:

- 1- Cut a strip of each color from the dry chromatogram
- 2- Slip it in a little substance to dissolve the substance
- 3- Evaporate the solvent
- 4- So now the mixture is fully separated, you will end up with very tiny samples that you can analyse

2- Checking whether a substance is pure:

- 1- Draw a pencil line on the paper, put a drop of a concentrated solution (X) on the line, and let it dry. Its position on the line is called the origin.
 - 2- Stand the paper in a little solvent, with the solvent level below the pencil line. The solvent rises and X begins to dissolve.
 - 3- After some time, if X separates into two blobs, the X is impure. If there was one blob only, X is pure
- * The line is drawn in pencil not pen because ink on the paper might dissolve.

3- Identifying substances in a mixture:

- 1- Prepare concentrated solutions of X (unknown), [A, B, C, D,] Known, in propanone. Place a spot of each along a pencil line on a chromatography paper, and label them.
- 2- Stand the paper in little propanone, in a covered glass tank. The solvent rises up the paper, when it is near the top, remove the paper.
- 3- Compare the spots where X has separated with the known substances.